

PROJECT EVIDENCE DOSSIER · ENGLISH

Shared mini-biorefinery system · Copey de Dota

Systematization of the digital plant guide developed for Impulsa CR: invite-only access, hygienic layout for a 4.0 × 5.5 m (22 m²) shed, interactive drawings, works checklist, permits, GMP, water, solar power and equipment — as published on Rural Commerce.

22 m² · 4.0 × 5.5 m

Frutalcoop · Copey de Dota

Invite-gated guide

ruralcommerceglobal.com

01 · PURPOSE

Purpose of this dossier

This document consolidates, in English, the agro-industrial plant system designed and published for the Impulsa CR project: a shared mini-biorefinery beside the collection center in Copey de Dota (Municipality of Dota, Costa Rica), operated with Frutalcoop.

It is intended as **project evidence** — what Rural Commerce developed in the digital guide — not as a substitute for stamped engineering drawings or municipal filings.

What was developed

A multilingual (ES / PT-BR / EN), invite-only web guide integrated into the Impulsa CR area of ruralcommerceglobal.com:

Guide modules

- Home / overview
- Works checklist (browser-saved progress)
- Interactive drawings (9 views)
- Legal / permits
- GMP (good manufacturing practices)
- Water, power & wastewater
- Equipment & meters

Drawing views inside “Plantas”

- Original cooperative sketch + review
- Zoning (interactive zones)
- Process flow (7 steps)
- Equipment plan
- Interactive 3D view
- Materials & partitions
- Electrical & plumbing
- Water networks
- Solar / energy

Design language.

Same visual system as the Impulsa CR / Rural Commerce project surfaces: Lexend type, navy (#071F5E), teal (#009179), soft paper background, pill navigation, and rounded content cards.

Invite-only access model

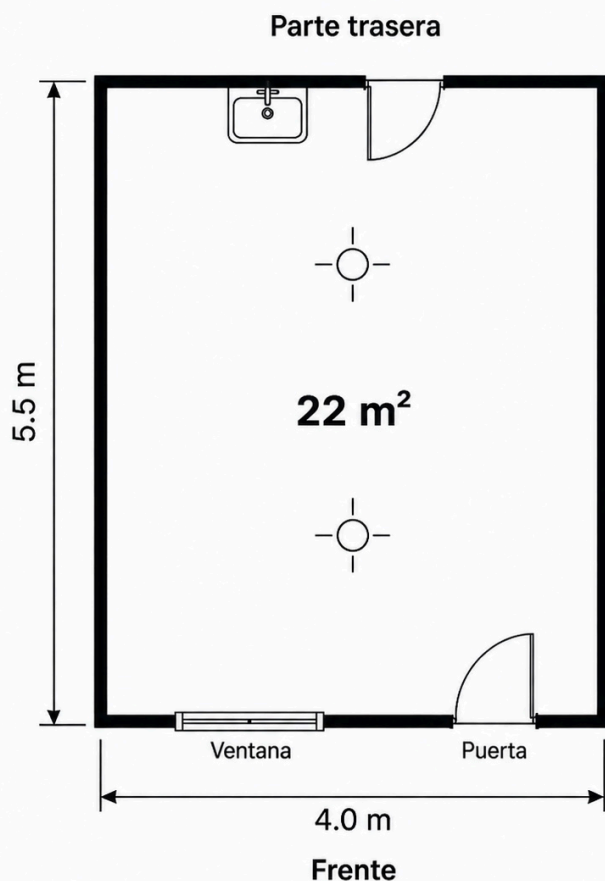
The Copey mini-biorefinery guide does **not** open with beneficiary login or the intranet. Entry requires an invite code issued by the Rural Commerce team (WhatsApp or email).

- Invite gate: /en/impulsacr/convite
- Plant shell after redeem: /en/impulsacr/planta...
- Demo code used with the cooperative for walkthroughs: **COPEY-22M2**

Facility overview

Shed footprint **4.0 × 5.5 m (22 m²)** beside the collection center in Copey de Dota. Product and people move: receive → wash → peel/cut → transform → pack → store → exit.

- **P1** — front door: people in / finished goods out
- **P2** — service door: raw material from the collection center
- **V1** — front window · **V2** — side window with insect screen



Original sketch supplied by the cooperative (reviewed in the “Original” drawing tab).

05 · HYGIENE LOGIC

Hygienic zoning (one room, clear flow)

In 22 m² the shed is one continuous room. Floor marks and process order create hygiene — not separate changing rooms between every strip.

Dirty · receiving

Collection-center side: crates, raw material and waste. Enters by the back door. No packing here.

Wet · wash, cut, transform

Sink, peel/cut table, then dry, mill or cook by recipe. Sloped floor. Steam must not blow toward packing.

Clean · pack & store

Packing, finished-goods shelf and exit. Enter with washed hands. Floor +1 cm vs wet zone.

People clothing

Staff put on hairnet / apron / gloves once at the front entry, wash hands at the pedal sink, then work the dirty→clean line in the same process attire. Floor tape (e.g. orange between receiving and wash) is a visual hygiene cue, not a door requiring a clothing change.

06 · PROCESS

Seven-step product flow

01 Receive

02 Wash

03 Peel & cut

04 Transform

05 Pack

06 Store

07 Exit

1. **Receive** — Back door from the collection center. Crates, not the floor.
2. **Wash** — Back sink, potable water, drain D1.
3. **Peel and cut** — Work table. Already washed product.
4. **Transform** — Dry, mill or cook: the recipe chooses. Same zone.
5. **Pack** — Clean zone. Lot, date, moisture.
6. **Store** — Finished-goods shelf, apart from raw material.
7. **Exit** — Product out the front door.

Rule: nothing dirty crosses the packing table. Never reverse the path on the same tray.

Zones in detail

Zone	Role	Size	Do / don't
Receiving	Raw material from collection center (P2)	1.95 × 1.20 m	Plastic crates, steel table · no finished goods
Wash	Wash produce & utensils	2.05 × 1.20 m	Steel sink, potable, D1 · not a staff handwash
Peel & cut	Peel, chop, portion	4.00 × 1.05 m	Steel table, T2 · no packing / no dirty crates
Transformation	Dry, mill or cook	4.00 × 1.35 m	Cabinet T4, mill T3, gas stove + extractor · steam away from packing
Packing	Clean dry packing	2.35 × 0.95 m	Table, sealer, scale · floor +1 cm
Storage	Packed product	1.65 × 0.95 m	Clean shelf · away from raw material
Exit / entry	Staff in · goods out (P1)	4.00 × 0.95 m	Pedal handwash, apron wall, mat

Separations

- Dirty / wet (back) vs clean (front): packing and storage sit 1 cm higher.
- 1.20 m screen between transformation and packing; 50 cm clear around each machine.
- Peel and cut has its own strip — not a corner of the sink.
- Drying, milling and cooking share transformation; the recipe picks the technique.

Drawings suite (systematized plants)

All views share doors/windows (P1, P2, V1, V2) and the same 4.0 × 5.5 m envelope.

Original

Cooperative sketch + keep / drop review against hygienic flow (stove not in packing; dehydrator indoors; six covered lights; slope and drains).

Zoning

Interactive zone chips with role, do/don't and separation notes for each strip.

Process

Numbered path on the plan plus the seven-step narrative.

Equipment / 3D

Plan symbols and an interactive 3D shed: orbit, zoom, pan; technique filters dim unused transform machines; click syncs the equipment selector.

Materials

Partitions, raised packing floor, screen line toward transformation.

Technical

Receptacles (NEMA, GFCI, circuits), lighting points, drains D1/D2, panel TAB and solar feed C7.

Water

Blue potable network vs green rain network; first flush 40 L; 1,000 L tank; grease trap.

Energy

Three 625 W modules (1.88 kWp), ~3 kW hybrid inverter at C7, no batteries in base design; ICE as backup.

Works checklist (build-now items)

Urgent items highlighted in the live guide while the shed is being built:

- **Sloped floor** 1–2% toward drains (not flat).
- **Two floor drains** with pest screens (wet zone + near service sink).
- **Second handwash** at the front door (hands-free + paper).
- **Six covered luminaires** (IP65 class), not two bare bulbs.
- **Conduit to the roof** for solar before ceilings close.

Additional checklist items: back-wall process sink, grease/solids traps, dry-zone electrical panel, potable water without cross-connection, washable surfaces, screened window, door sweeps, floor zone marks, chemicals niche, lidded waste exiting via the dirty route.

Dota and Health permits

Municipality of Dota

Land use, building permit if required, municipal business license. Confirm small-scale food industry is allowed beside the collection center.

Los Santos Health Area

Operating sanitary permit (PSF) before operations. Risk group under Decree 43432-S; drawings, sworn statement, Article 8 preconditions; processed-food GMP inspection sheet.

ICE and ASADA / AyA

Real electrical load; consumption meter; generation meter if solar; potable water with chlorine records if treated on site.

Wastewater & environment

Pretreatment before discharge (grease trap, solids basket). Decree 33601-S-MINAE; SETENA only if a D1 applies — 22 m² is not an automatic waiver.

Good manufacturing practices

People flow

Front door → pedal handwash and aprons → process. Product enters by the back door. Do not move from dirty to packing in the same apron or boots.

Product flow

Collection center → receiving → wash → peel/cut → transform → pack → store → exit.

Hygiene

Soap, short nails, hairnet, washable apron. Back sink = process; front sink = entry.

Records

Daily cleaning, cook time/temperature, product moisture, water chlorine if dosed, incoming lots, trap cleaning.

Pests

Screens, door sweeps, no food on the floor, no standing water.

Inspection focus

Slope, water, hands at entry, wastewater, lighting; packing/storage higher; stove stays in transformation.

Water & wastewater

1 · Blue (potable)

AyA/ASADA for food, final rinse and handwash. Never crossed with rain (RTCA 5.3).

2 · Green (rain)

22 m² roof → first flush 40 L → 1,000 L tank. Floors, dirty crates, outdoor sink, irrigation. Filtered, not potable.

3 · Out

Sinks and floors → drains D1/D2 → solids basket → grease trap 50–100 L → authorized discharge. Never to the creek untreated.

Indicative capture: ~39,000 L/year possible from the roof at local rainfall; day-to-day storage is the 1,000 L tank.

Solar & electricity

Design on the published guide: **three 625 W modules (1.88 kWp)** on the 22 m² roof, **~3 kW hybrid inverter** at panel circuit C7 (DC→AC), **no batteries** in the base design. ICE 60 A service with bidirectional meter as backup for clouds/night. The industrial stove is gas.

Load	W	h/day	kWh	Source (design)
Electric dehydrator	1200	5	6.0	Solar (day)
Mill	1100	1	1.1	Solar (day)
6 LEDs	108	6	0.6	Solar (day)
Sealer and table	300	2	0.6	Solar (day)
Rainwater pump	250	1	0.3	Solar (day)

Indicative daytime yield ~5.9 kWh/day in Dota conditions for the three-module array. Separate circuits for lights, process receptacles, pump and solar reserve; grounding; do not feed the plant from an extension in the collection center.

Equipment & meters

Item	Specification (guide)
Process sink	Stainless, two bowls; potable; drain to D1
Receiving table	Steel; plastic crates; raw material never on the floor
Cutting table	Steel; peel/chop/portion; T2
Electric dehydrator	Indoor tray cabinet; 1.2 kW; T4 120 V
Mill	1–1.5 HP; dedicated T3 240 V 20 A
Industrial stove	Four burners; outdoor extractor; fat to grease trap
Moisture extractor	Duct outdoors with screen; air not toward packing
Packing table / sealer	540 lux; T6; lot and date
Finished-goods shelf	Apart from raw material
Apron wall / staff handwash	Entry barrier; pedal/knee/sensor; RTCA 5.4.3
Panel TAB	12 spaces, 60 A, 120/240 V, inverter C7
Outdoor sink / rain tank	Crates & boots; green network; 1,000 L tank

Project follow-up meters (water, electricity, product humidity): support Impulsa CR coaching; they do not replace the ICE fiscal meter.

Live evidence & URLs

Published on the Rural Commerce production site:

- Project hub: <https://ruralcommerceglobal.com/en/impulsacr>
- Invite gate: <https://ruralcommerceglobal.com/en/impulsacr/convite>
- Plant guide (after invite): <https://ruralcommerceglobal.com/en/impulsacr/planta>
- Drawings: .../[planta/croquis](#)
- Works · Legal · GMP · Services · Equipment: sibling routes under .../[planta/](#)

Also available in Spanish and Portuguese (/es/, /pt-BR/) with the same structure.

How to use this file.

Attach this HTML or a PDF printout to project reports as evidence of the digital agro-industry plant system delivered by Rural Commerce for Impulsa CR. Keep engineering stamp sets and municipal submissions as separate formal packages.